Nosocomial infections

Dr. Shireen Ahmed Dzayee
Department: Microbiology, Physiology, and Genetics
College of Medicine
Hawler Medical University

1st Stage

LG 5

List of contents

Nosocomial infections

- Nosocomial infections, also known as hospital-acquired infections (HAI) defined as:
- Those infections presenting during hospitalisation (at least 2 days after admission) by a patient who was not incubating that infection upon admission.

Sources of infection

Endogenous:

- These are infections caused by microorganisms that are part of the host's normal flora.

Exogenous :

These infections are caused by microorganisms acquired from other humans, animals or the environment.

Sources of infection

Humans: infections are commonly acquired from other humans, who may be:

a. Infected

b. Carry a microorganism asymptomatically; carriers may be convalescent (i.e. recovered from the infection, but continuing to excrete the microorganism, e.g. typhoid carriage in the gall bladder);

c. Healthy carriers who have not had an infection (e.g. a third of the population persistently carry Staphylococcus aureus in the nose).

Sources of infection

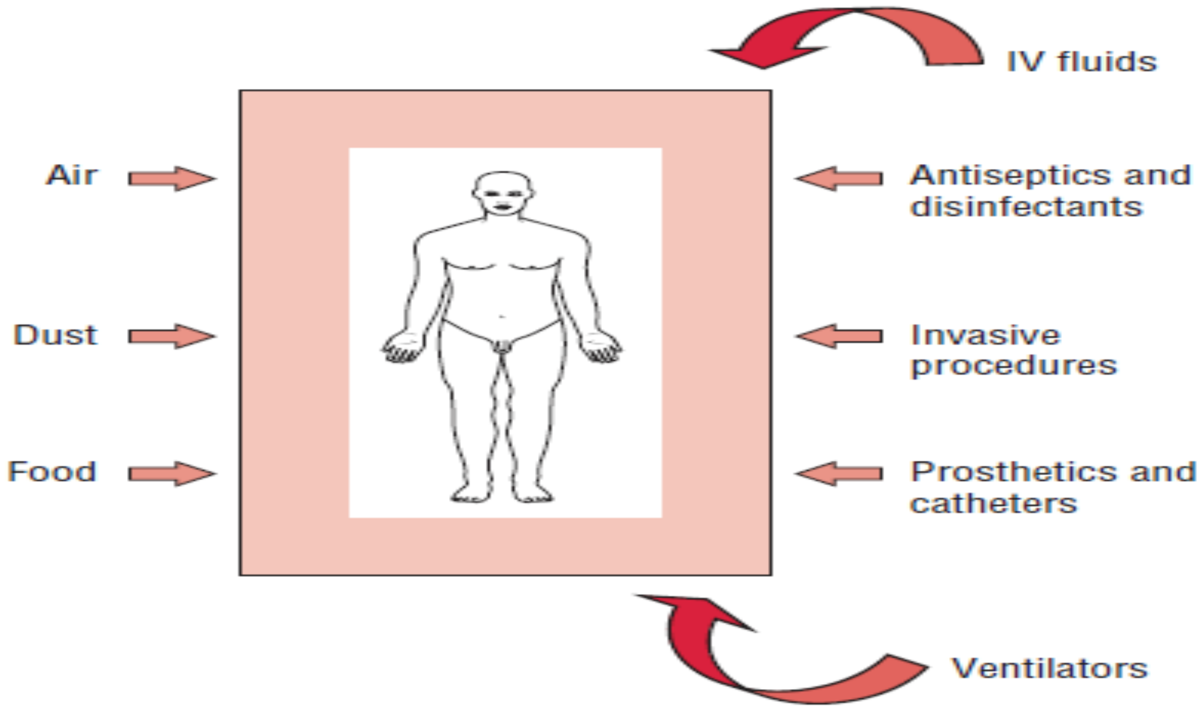
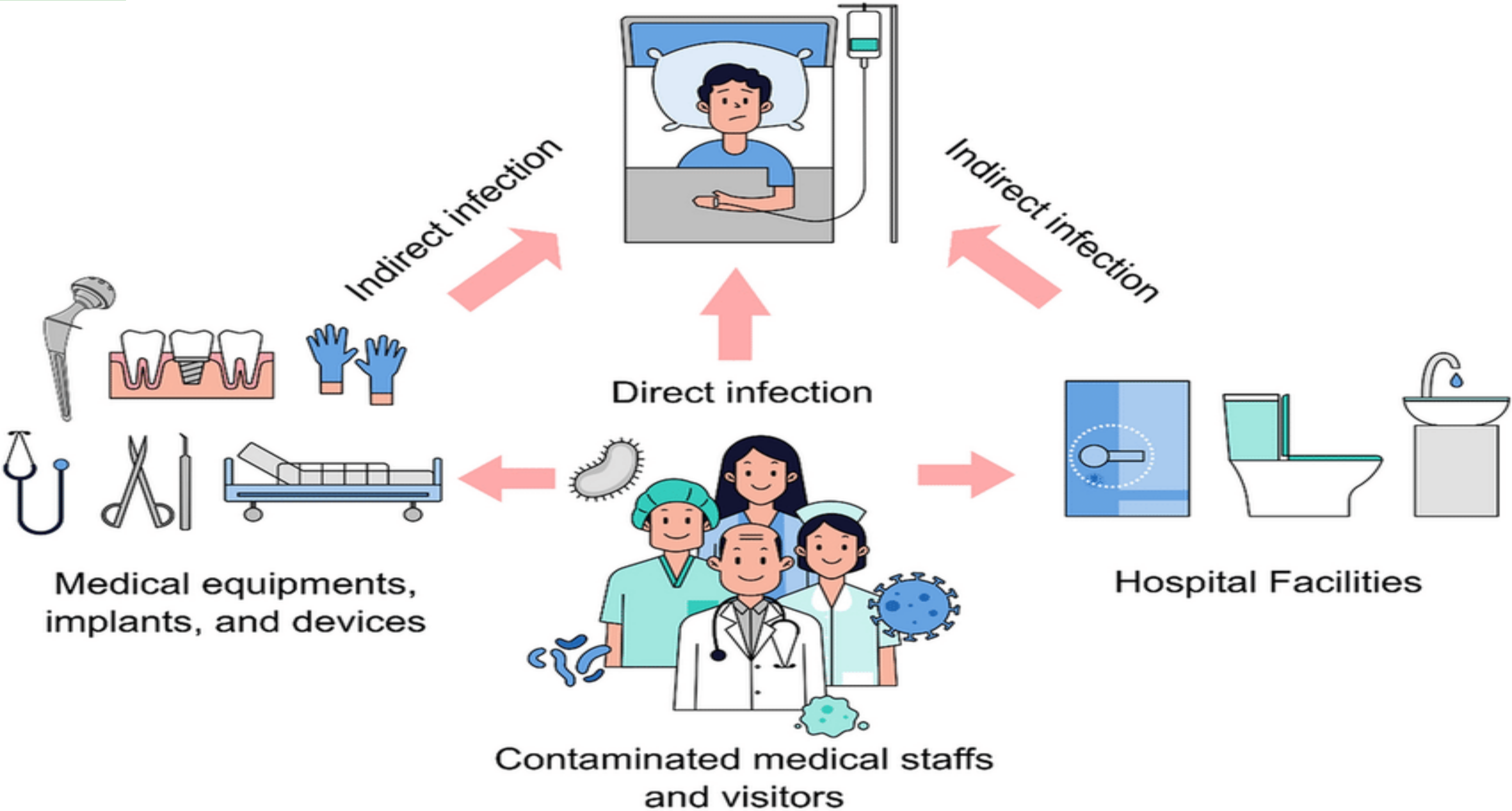


FIG. 77-1. Environmental sources of nosocomial infections.

Sources of infection and Transmission of infections

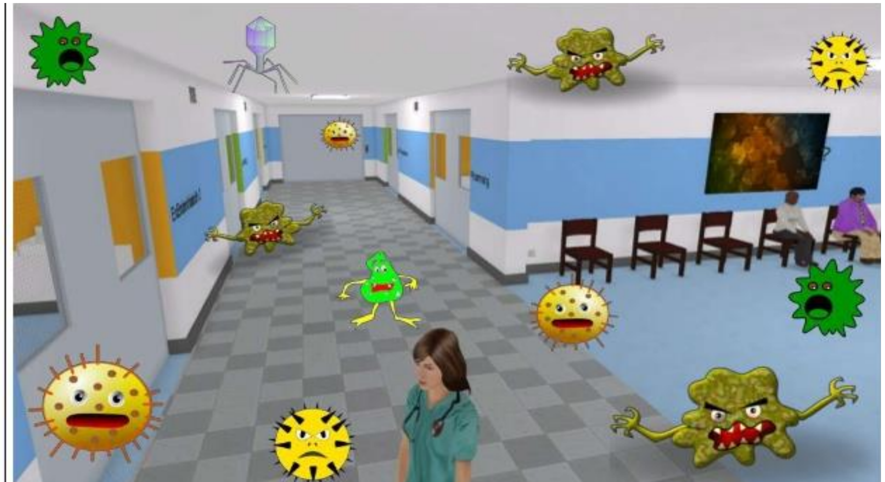


Transmission of infections

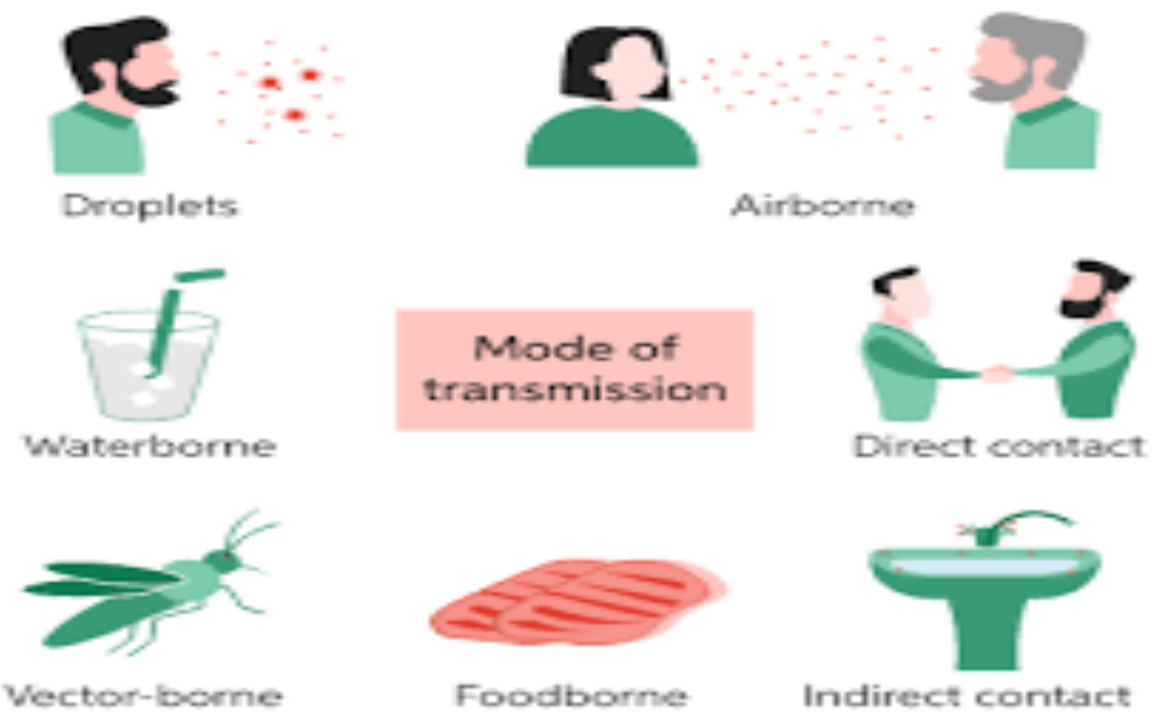
- **Direct transmission:**
  - presence of a source of infection and a susceptible individual (immunocompromised)
  - contact (e.g., kissing/sexual contact);
  - essentially transmission by the hands of health care personnel;
  - in newborns - eye infection (direct contact with the mucous membrane of the vagina);
  - droplet infection;
  - alimentary route - preparation of milk food in the neonatal unit.

Transmission of infections

- **Indirect transmission**
- depends on the ability of the microorganism to **survive** outside the host body;
- the existence of a suitable medium in which the microorganism multiplies and with the help of which the infection is transmitted



Transmission of infections



Causes nosocomial infections

- Bacteria, fungus, and viruses can cause HAIs.
- Bacteria alone cause about 90% of these cases.

- Many people have compromised immune systems during their hospital stay, so they’re more likely to contract an infection.

Causes nosocomial infections

- Bacteria, fungi, and viruses spread mainly through person-to-person contact.
- This includes unclean hands, and medical instruments such as catheters, respiratory machines, and other hospital tools.
- HAIs cases also increase when there's excessive and improper use of antibiotics. This can lead to bacteria that are resistant to multiple antibiotics.

Causes nosocomial infections

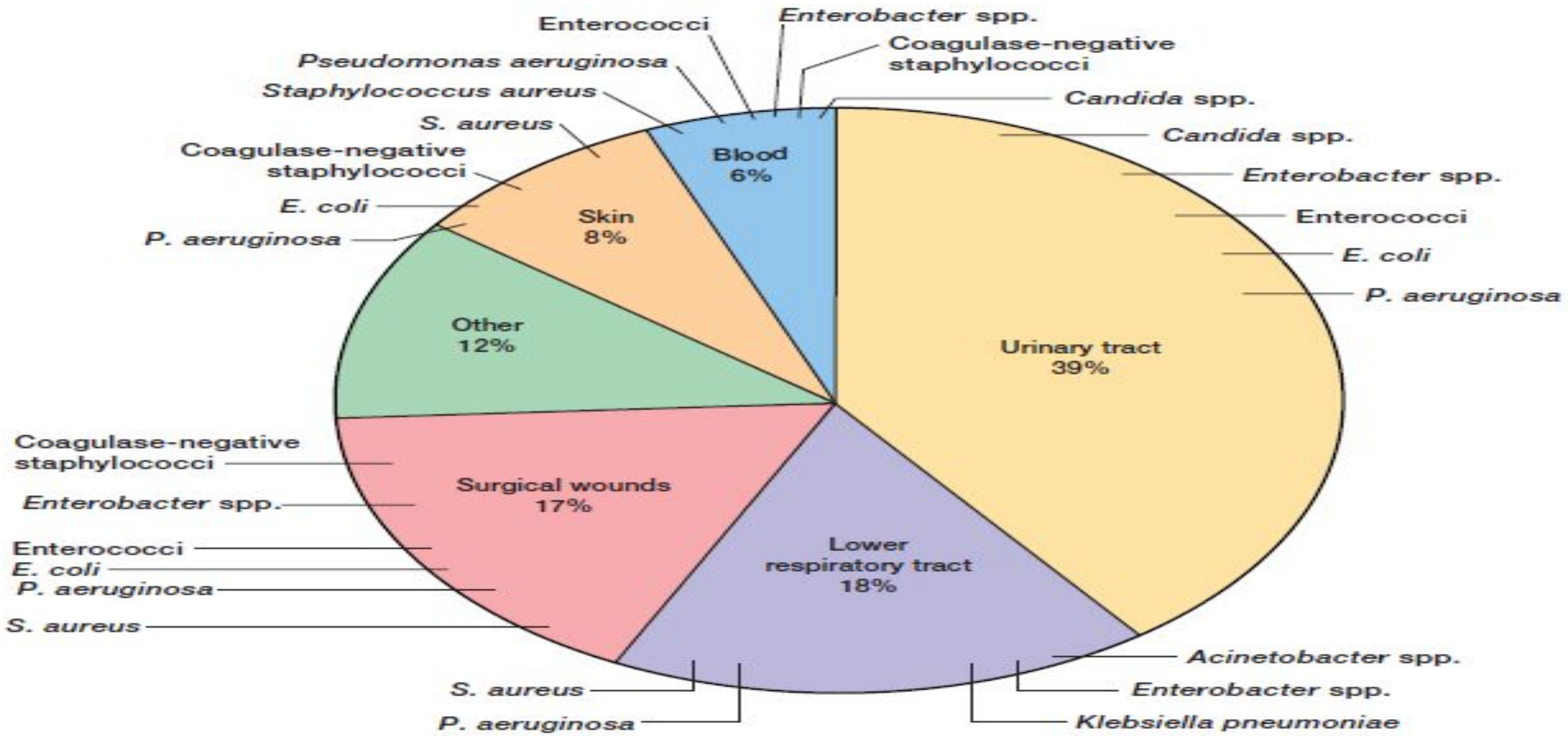
Bacteria	Infection type
Staphylococcus aureus (*S. aureus*)	blood
Escherichia coli (*E. coli*)	UTI
Enterococci	blood, UTI, wound
Pseudomonas aeruginosa (*P. aeruginosa*)	kidney, UTI, respiratory

Types of Nosocomial Infections:

Most Common Types of Nosocomial Infections

1. Urinary tract infections.
2. Surgical site infections.
3. Lower respiratory Tract infections (primarily pneumonia).
4. Bloodstream infections (Septicaemia)

Relative frequency by body site . These data are from the National Nosocomial Infections Surveillance, which is conducted by the Centers for Disease Control and Prevention (CDC).
***Pseudomonas aeruginosa* (*P. aeruginosa*) *Escherichia coli* (*E. coli*) *Staphylococcus aureus* (*S. aureus*)**



Nosocomial infections risk of infection

Increased risk of infection is related to:

1) prolonged hospitalisation,
2) intensive care,
3) the use of invasive prosthetic devices (e.g. intravenous catheters, urinary catheters, endotracheal tubes) and
4) immunocompromised patients, including those with impaired local host defences (e.g. burns, trauma).
5) The widespread use of antimicrobials in hospitals results in the selection of antimicrobial-resistant microorganisms in the hospital environment

Symptoms of nosocomial infections

The infection must occur:

- up to 48 hours after hospital admission
- up to 3 days after discharge
- up to 30 days after an operation
- in a healthcare facility when someone was admitted for reasons other than the infection

Symptoms of nosocomial infections

- discharge from a wound
- fever
- cough, shortness of breathing
- burning with urination or difficulty urinating
- headache
- nausea, vomiting, diarrhea
- People who develop new symptoms during their stay may also experience pain and irritation at the infection site. Many will experience visible symptoms.

Symptoms of nosocomial infections

Diagnosis of Nosocomial infection

- The specimens are usually sampled from possible sources of infections, such as environment, inanimate objects, water, air, food, hospital personnel, etc.

- Routine bacteriological methods, such as
  - Direct demonstration of microorganisms in specimens by microscope.
  - Isolation by culture.
  - Testing bacterial isolates for their antibiotics sensitivity pattern.

Prevention and control of Nosocomial infection

- Maintaining proper personal hygiene and hand washing
- sterilisation of hospital equipment
- providing clean and sanitary environment
- existence of infection control team
- regular, close observation of high-risk units .e.g intensive care
- development of policies on areas such as isolation, disinfection and antibiotic usage

Prevention

HAWLER MEDICAL UNIVERSITY

References

- Patrick R. Murray, Ken S. Rosenthal and Michael A. Pfaller. 2021. Medical Microbiology. 9th edition. Elsevier.
- Nosocomial infections :Introduction, source, control and prevention; 2020 by Vivek Kumar
- Richard V. Goering, Hazel M. Dockrell, Mark Zuckerman, Peter L. Chiodini. 2019. Mims' Medical Microbiology and Immunology. Sixth edition. Elsevier.